

# Does Economic Analysis Improve the Quality of Foreign Assistance?

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*The World Bank undertakes an annual expenditure of around \$60 million on country-specific economic analysis and advice for its member developing countries. What is the impact of this economic and sector work on the quality of World Bank lending? It would be useful to know whether past analytical work has generated measurable economic benefits that would justify its continued provision in an environment of increasingly scarce resources.*

*This article sets out an idealized model of decisionmaking in which a country manager makes a broad allocation of resources between lending services and economic and sector work. Given that decision, the task manager for each project makes project-specific decisions with respect to the allocation of resources between preparation and supervision. The analysis indicates that economic and sector work has a significant positive impact on the quality of World Bank loans. The results provide clear evidence of underinvestment in economic and sector work. And the analysis shows that resources could be switched from preparation and supervision to economic and sector work to the benefit of both the quality of programs and level of disbursements.*

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What is the benefit of the millions of dollars of foreign assistance provided to developing countries in the form of economic analysis and advice? This is a difficult question to answer. In the absence of a clear market test, there is no simple way to value the contribution of analytical work; most of it is provided free of charge or embedded in an investment or program loan. And the impact of economic analysis on some ultimate objective, such as development or poverty reduction, is almost impossible to identify because it is only one of many factors that determine outcomes.

Yet, it would be useful to know whether past analytical work has generated measurable economic benefits that would justify its continued provision in an environment of increasingly scarce resources. Moreover, some analysts claim that with improvements in the access of developing countries to nonconcessional sources of finance, the comparative advantage of international institutions such as the World Bank will shift toward the provision of analytical services (see, for

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example, Rodrik 1995). But little empirical evidence exists that could be used to substantiate such a claim or to provide guidance on how and where to focus such services in the future.

The issue tackled in this article faces the same problems that confront efforts to evaluate research more generally. The literature on this topic (including this article) takes the view that spending on research is an investment that should produce measurable economic returns. The difficulty, of course, arises with the word measurable. Traditionally, economists have based their analysis on the production function approach or the consumer surplus approach (Averch 1994). Both approaches rely on market outcomes to provide measures of the benefits of research. Often, however, it is not possible to use market outcomes. Researchers frequently try to create more basic knowledge that, like all public goods, proves difficult to value. This happens to be true in our case. Economic studies and policy advice create and disseminate knowledge, the ultimate impact of which can be far-reaching but is usually difficult to trace.

The analysis could focus more directly on the immediate outputs of research, meaning the reports generated or articles published (Wise and Agranoff 1991). For example, the National Science Foundation identifies a set of indicators that includes publications, citations, data archiving and sharing, software development and sharing, and education and pedagogy (National Science Foundation 1995). The World Bank has also used a variant of this approach, specifically, the number of World Bank publications appearing in university reading lists (World Bank 1991). Apart from the incentive problems that these measures may create, they fail to provide any measure of the value of research. In short, the available literature on the evaluation of research does not provide much guidance on how to value the output from economic analysis and advice and hence how to estimate the return to such activity.

This article provides a partial answer to this question. It does not encompass all the benefits of economic analysis and advice. Foreign assistance provided in the form of analysis and advice attempts, among other things, to maintain a policy dialogue with the client country in order to provide analytical support for critical macroeconomic decisions, examine the steps needed to generate and maintain a policy and regulatory environment that promotes private sector activity, provide analytical inputs to the design of sectoral strategies, and build local capacity for policy analysis and discussion of key policy issues in the broader context of civil society. In addition, economic analysis and advice often provide a critical underpinning for the design of individual projects and lending programs. This last aspect is the focus of this article. As such, our results will be a minimum estimate of the full benefits from this form of foreign assistance.

To our knowledge, the only data set providing measures of resources spent on analytical services as well as measures of the benefits of lending is for the World Bank. Thus the purpose of the article is to assess the impact of the economic and sector work (ESW) provided by the World Bank on the quality of World Bank lending. Although data availability forces us to focus on the World Bank's ESW,

this is not insignificant either in coverage or in quantity. ESW is undertaken on all developing countries that are members of the World Bank, numbering well over 100 countries (see World Bank 1997). The work undertaken on each country is specific to that country but could include the following: a country economic memorandum (a comprehensive account of economic performance and prospects) and more topic-focused reports such as poverty assessments, public expenditure reviews, labor market studies, as well as a wide range of sectoral studies. In general, the more aggregate studies underpin the World Bank's policy advice, while the sectoral studies (reviews of the transport sector, health and education sector reports, and so on) provide the foundation for lending operations in those sectors. Undertaking this range and magnitude of work is expensive. Since 1975, the World Bank has used about 22,000 staff-weeks a year on ESW. This amounts to an annual expenditure of around \$60 million on country-specific economic analysis and advice. This almost certainly makes the World Bank one of the largest sources of such analysis for the developing world.

In this article, we address three main questions. First, does ESW improve the quality (variously defined) of World Bank loans? Second, has the World Bank underinvested in ESW relative to lending services? Third, is there a trade-off between quality and volume of lending?

We set out an idealized model of decisionmaking in which managers are assumed to be concerned exclusively with development impact. We assume that managers have sufficiently long planning horizons and sufficient information to be able to allocate resources broadly in line with their objective. Of course, in reality managers may be influenced by factors other than development impact, they may have relatively short time horizons, and the information relevant to successful decisionmaking at best may be only partial. The intention then is not to treat the idealized model as a description of reality, but to see how close reality approximates the ideal given all the factors that inevitably beset decisionmaking in the real world. Where we observe departures from this ideal, we try to explain the difference.

Our model of decisionmaking operates at two levels. We envisage a country manager who makes a broad allocation of resources between lending services and ESW. Given that decision, the task manager for each project makes project-specific decisions with respect to the allocation of resources between preparation and supervision. So at the country level, decisions are assumed to be made by a country manager who can allocate staff time freely between lending services and ESW. In this model, therefore, ESW is *endogenous*. The country manager is assumed to maximize development impact subject to the overall availability of staff resources. ESW contributes to development impact through its contribution to the quality of the countrywide lending program (defined as the total number of projects in a country approved in a given year) and through its contribution to policy formulation more generally. Testing this model allows us to address all three questions posed above. The model also yields (implicitly) the shadow price of a unit of staff resources. This price enters the decisionmaking problem of the

task manager responsible for a single project, the second level of decisionmaking that we examine. At this level, we assume that ESW is *exogenous* and that the manager's objective is focused more narrowly on maximizing the quality of the project. This model provides additional evidence on the first two questions posed above.

Section I places the study in the context of other work on foreign assistance and highlights how our data constitute an improvement over those used in earlier studies. Section II discusses the conceptual framework and estimating equations. Section III describes the data used in the study and provides descriptive statistics. Section IV presents our empirical results. Concluding remarks are presented in section V.

## I. RELATION TO THE LITERATURE

Research related to foreign aid falls into three broad categories. Game-theoretic models focus on the negotiation and implementation of conditionality and on the credibility and commitment problems that arise if the donor or the recipient behave strategically (Mosley, Harrigan, and Teye 1995 and Svensson 1996). Empirical models analyze how donors allocate aid across recipient countries and how aid may affect the decisionmaking and economic performance of recipient countries (see Burnside and Dollar 1996; Pack and Pack 1990, 1993; Feyzioglu, Swaroop, and Zhu 1995; and Trumbull and Wall 1994). Descriptive or quantitative efforts analyze the factors that affect the performance of individual donor-financed operations and countrywide programs (see, for example, Claessens, Pohl, and Quian 1996, Arias 1994, Cassen and Associates 1994, Isham and Kaufman 1995, Kilby 1994, and Lele 1992). This article is most closely related to the third category. It attempts to identify the quantitative contribution of economic analysis to the quality of individual investment projects and the size of overall resource flows. Unfortunately, data availability limits the scope of the study to assistance provided by the World Bank.

The impact of economic analysis on the success of World Bank projects has been investigated in two other studies with conflicting conclusions. Arias (1994) defines ESW as the number of World Bank reports on the country published in the three years preceding project approval. He fails to find a significant effect of ESW on project quality. This finding is based on an analysis of all projects that were *evaluated* between 1991 and 1994. Analyzing projects *approved* between 1991 and 1994, and defining ESW the same way, Schneider (1995) finds that ESW has a significant quality-enhancing impact. These contrasting results are a further motivation for the present exercise.

Although the basic issues to be investigated are the same, this article differs from those studies. We try to cast the problem in terms of a more explicit decisionmaking framework. And both the quality and coverage of data are significantly expanded. Using data on input measured in staff-weeks of analytical services rather than the number of written reports allows us to reduce measure-

ment errors that may have affected other studies. And relying on all available information in the World Bank's databases facilitates the inclusion of projects that were approved after 1985.

## II. CONCEPTUAL FRAMEWORK

In this section we describe the processes governing the allocation of resources to different activities within the World Bank in the form of two decisionmaking models. We derive the equations for estimation and specify empirically testable hypotheses. We start with the model of a country manager responsible for all activities in a given country and then turn to that of a task manager at the project level.

### *Model 1: Country Manager*

We assume that the country manager's objective is to maximize a weighted function of lending program quality,  $Q$ , and policy impact,  $P$ , subject to the total staff time,  $T$ , available for ESW and all lending services—both investment lending, INV, and adjustment operations, ADJ. Formally, the country manager solves the following problem:

$$\text{Max } L = U [Q(\text{ESW}, \text{INV} \dots); P(\text{ESW}, \text{ADJ} \dots)] - \lambda (\text{ESW} + \text{INV} + \text{ADJ} - T).$$

The solution to this problem will yield demand functions for ESW, INV, and ADJ contingent on the overall budget constraint as well as various exogenous factors, such as country-level policies and institutional factors.

In principle, these demand functions, together with the exogenous constraints, could be used to solve for the optimum values of  $Q$  and  $P$ . Empirically, however, it is impossible to distinguish between the parameters of the production functions for lending quality and policy impact ( $Q$  and  $P$ ) and the relative weights that the decisionmaker assigns to each of these outcomes. Nor do we have a variable to measure policy impact. To answer the questions posed earlier, therefore, we estimate a reduced-form equation for  $Q^*$ , the optimum value of lending quality. Using a linear approximation, we estimate the following equation.

$$(1) \quad Q_{jt}^* = \mu_0 + \mu_1 \text{ESW}_{jt} + \mu_2 \text{INV}_{jt} + \mu_3 \text{POL}_{jt} + v_{jt}$$

where  $j$  and  $t$  denote country and year, respectively. We include variables to capture policies (POL) and may adjust for other unobservable country-specific factors by using dummies.

While the observed value of ESW can be taken to be a reasonably accurate estimate of the decisionmaker's choice, this is not so for INV. Instead, what is available in our data is the amount of resources that was actually spent on INV (preparation and supervision). This amount may differ from the country manager's plans if emergencies or external events cause management to allocate additional funds to redesigning projects during preparation or to restructuring the project during implementation. Furthermore, because task managers have

no incentive to reveal negative information about individual projects that is not available to the country manager, or because of a general desire not to cancel projects, more time may be systematically allocated to bad projects in practice than was intended by the country manager. To deal with this, we instrument for INV using anticipated project age, past supervision, and past preparation requirements as well as past project success in the country.

Equation 1 allows us to examine all three questions of interest. First, does ESW improve the quality of World Bank loans? We expect that, if ESW contributes to the quality of the lending program,  $\mu_1 > 0$  (hypothesis 1).

Second, has the World Bank underinvested in ESW relative to lending services? Resources allocated to ESW compete with resources for INV, both of which should in principle improve project quality. According to our definition, the lending program consists of all the projects approved in a given year; this would imply some kind of intertemporal reallocation, that is, higher expenditures on current ESW would be rewarded by lower spending on INV in the future. ESW also contributes to policy impact. Through its impact on economywide and sector-specific policies and institutions, economic and sector work generates benefits beyond the projects supported by the World Bank. Therefore, if staff resources are allocated to maximize the contribution to development, the marginal contribution of ESW to the quality of the lending program plus its marginal contribution to policy impact should equal that of INV. As long as the impact of ESW on policy is positive, it follows that the marginal contribution of ESW to project quality should be less than that of INV. That is, if the World Bank has systematically maximized development impact, we should expect  $\mu_1 < \mu_2$  (hypothesis 2).

Third, is there a trade-off between the quality and volume of lending? If the idealized model is supported by the data, we can conclude that managers have not compromised quality to increase the volume of lending. If the model is not supported by the data, then such a trade-off may underlie the observed outcome. We take up this issue in section IV.

### *Model 2: Task Manager*

We now turn to the decision facing a task manager of a single project. The task manager's decision differs from that of the country manager in three respects. First, ESW is taken as exogenous. That is, we can reasonably assume that decisions about ESW (macroeconomic analysis as well as sector-specific analysis) are made by senior management rather than by task managers associated with individual projects. Moreover, we assume that those decisions occur well before individuals are assigned to the preparation of specific projects. Second, the task manager of an investment project has little control over economywide policies. We assume, therefore, that the manager at the project level focuses exclusively on project quality. To the extent that the project design includes some changes in policy, we assume that the task manager's primary impact is on the quality of the project itself. And third, we divide INV into preparation time (PRP) and supervision time (SPN), both of which are assumed to contribute to project quality. We

treat PRP and SPN as substitutes because a well-prepared project reduces the need for subsequent supervision, and more supervision is required to rescue a badly prepared project. The task manager chooses the level of PRP and SPN to maximize the quality of the project subject to the given shadow price of staff time. In the event that this price is high, the project task manager is obliged to cut corners and economize on staff input. In the event that the price is low, the manager is able to prepare and supervise the project more thoroughly.

In general, the task manager's optimization problem is:

$$\text{Max } L = Q(\text{PRP}, \text{SPN}; \text{ESW}, \text{POL}, C, S \dots) - \lambda (\text{PRP} + \text{SPN})$$

where  $\lambda$  is the shadow value of time derived from the country manager's optimal solution, and  $C$  and  $S$  represent country-specific and sector-specific factors, respectively. The solution to this problem yields estimable demand functions for PRP and SPN in terms of the exogenous parameters. We use linear approximations and estimate the following two equations where both PRP and SPN are normalized by loan size:

$$(2) \quad \text{PRP}_{ijt}^* = \beta_0 + \beta_1 \text{ESW}_{jt} + \beta_2 \text{PPRP}_{jt} + \beta_3 \text{PQUAL}_{jt} + \beta_4 \text{POL}_{jt} + \beta_5 S_i + v_{ijt}$$

and

$$(3) \quad \text{SPN}_{ijt}^* = \gamma_0 + \gamma_1 \text{ESW}_{jt} + \gamma_2 \text{LOAN}_{jt} + \gamma_3 \text{PQUAL}_{jt} + \gamma_4 \text{PSPN}_{jt} + \gamma_5 \text{POL}_{jt} + \gamma_5 S_i + v_{ijt}$$

where in the preparation equation PPRP represents average past preparation requirements in the country, PQUAL is past quality of the country portfolio, and  $i$  denotes a project. In the supervision equation, LOAN is loan size, and PSPN is past supervision effort in the country and sector.

Equations 2 and 3 provide a partial test of the assumed decisionmaking model. The coefficients of greatest interest are  $\beta_1$  and  $\gamma_1$ . If  $\beta_1 < 0$ , average preparation requirements are reduced by the availability of prior ESW, implying that the task manager responds to the availability of ESW in a rational fashion consistent with the assumed decisionmaking process. Similarly, finding that  $\gamma_1 < 0$  provides empirical confirmation that managers respond to the higher quality of projects at approval (because of more prior ESW) by reducing the allocation of resources to supervision. We can also use these equations to test for several other possibilities. In particular, if only  $\beta_0 \neq 0$  or  $\gamma_0 \neq 0$  and  $\beta_i, \gamma_i = 0$  for all  $i$ , then preparation and supervision time are allocated merely on the basis of loan size. If only  $\beta_2$  (or  $\gamma_4$ ) are different from 0, preparation (supervision) are determined solely on the basis of past experience. We expect that unfavorable policies complicate both project preparation and implementation, that is,  $\beta_4 < 0$ ;  $\gamma_5 < 0$  (for policies where higher values indicate "better" policies). Similarly, higher past requirements for preparation and supervision at the country and sector level indicate the absence of institutional and administrative infrastructure, thus increasing the need for staff input ( $\beta_2 > 0$ ). Higher quality of the past country portfolio is expected to be associated with lower preparation requirements because existing infrastructure and talent can be drawn on ( $\beta_3 < 0$ ;  $\gamma_3 < 0$ ).

If the model posited is appropriate, lending program quality,  $Q$ , will depend on PRP, SPN, and ESW according to the following equation (where we again use a linear approximation):

$$(4) \quad Q_{ijt} = \mu_0 + \mu_1 \text{ESW}_{ijt} + \mu_2 \text{PRP}_{ijt} + \mu_3 \text{SPN}_{ijt} + \mu_4 \text{POL}_{ijt} + \mu_5 S_i + v_{ijt}.$$

Estimating this equation using the predicted values of PRP and SPN from equations 2 and 3 allows us to assess the marginal impact of ESW on project quality (hypothesis 1). In particular, if  $\mu_1 > 0$ , then we can conclude that ESW does improve the quality of projects. Indeed, using a quantitative indicator for project quality that can be expressed in dollar terms, we can measure the benefits of additional ESW and compare them with the nominal cost of staff input. In addition, we can use project-level data to investigate whether reallocating resources between ESW and preparation or supervision would improve project quality. Recall that because ESW makes a contribution to nonproject objectives, we expect  $\mu_1 < \mu_2$  and  $\mu_1 < \mu_3$  (hypothesis 2). If this hypothesis is rejected, we have further evidence that the country manager failed to achieve the appropriate allocation of resources across the three activities from the standpoint of development impact.

### III. MEASUREMENT ISSUES AND DESCRIPTIVE STATISTICS

Before discussing our empirical results, in this section we describe data definitions and sources and present descriptive statistics. We discuss the variables used to measure lending services and ESW, project success, and policies. Then we discuss bivariate relationships between the variables that will be of importance in the subsequent discussion.

#### *Lending Services and ESW*

Lending services are provided by the World Bank through projects that are associated with loans. In our analysis we consider a variety of project attributes such as the size of loan, the amount disbursed, and the amount of resources spent to prepare and supervise the project. We consider only loans that are either active or completed, thus discarding projects (about 10 percent of the total) that were dropped before becoming active. The amount of staff time (including long-term consultants) spent in preparation and supervision by project is available from the World Bank's management information system for all years following 1975. The variables for time spent in preparation and supervision are based on entries from the World Bank's time recording system, which classifies staff time devoted to specific activities by project and by task code. The only task codes encountered for lending projects are preparation, supervision, and preparation of the implementation completion report (which we have added to the time for supervision). Unless otherwise noted, preparation and supervision variables in all the regressions are normalized by loan size. That is, the short-hand terms preparation and supervision refer to staff-weeks of preparation or supervision time per \$1 million in loans.



ESW is defined as any use of resources by the World Bank (excluding research and other activities that are neither country-specific nor project-specific) that is not linked to the preparation or supervision of loans.<sup>1</sup> The main activities included in this category are macroeconomic and sector reports, public expenditure reviews, preparation of country assistance strategies, and donor coordination. We distinguish between general macroeconomic or multisectoral analysis and sector-specific ESW, with the latter disaggregated into seven sectors—agriculture and environment, education, health, finance, industry and energy, transport infrastructure and water, and public sector restructuring.<sup>2</sup> The variable for ESW is measured in weeks of staff time and is available on a consistent basis beginning in 1985. Because the variable for staff cost that is available in the World Bank database does not add any information (its correlation with staff-weeks is 0.82) and is available for significantly fewer observations, we measure ESW in staff-weeks throughout.

We link ESW, defined at the country or sector level, to specific lending projects by assuming that a project is affected by a simple average of the country and sector work undertaken during the four fiscal years preceding its approval by the World Bank's Board of Executive Directors. To account for the fact that macroeconomic work is likely to affect all projects in a given country while sector-specific ESW has a more limited impact, the ESW variable is defined as the sum of all macroeconomic ESW for the country in which the project is implemented and all sector-specific ESW for the sector in which the project falls using the seven sectors mentioned.

Table 1 illustrates that since 1975 the World Bank has on average spent about 53,000 staff-weeks (equivalent to about 1,260 staff-years) annually on supervision and preparation of loans, compared with about 22,000 staff-weeks invested in country-specific macroeconomic analysis (40 percent) or sector-related ESW (60 percent). Over time there have been considerable increases in the amount of resources devoted to supervision and sector-specific analysis, with much more modest increases in resources devoted to preparation of loans and to macroeconomic work. While total processing costs associated with lending stayed more or less constant at 3.2 staff-weeks per \$1 million in loans in 1977–92, the composition changed considerably in favor of more supervision and less preparation; the share of time spent in project preparation dropped from 68 to 58 percent. The share of ESW in total staff-weeks spent was consistently below 30 percent, and, even though a significant increase in spending on ESW occurred between 1989–92 and 1993–95, in absolute terms the increase was less than the expan-

1. Each year the World Bank spends about 26,000 staff-weeks on activities, mainly research and a few administrative functions, that are neither project-specific nor country-specific and that are therefore not included in our analysis.

2. Because the sectoral classification in the World Bank's management information system is quite poor (many sector-specific pieces are classified as general), we have reclassified the general category based on the task title. For example, an education sector review is placed into the education rather than the general category.

Table 1. *Evolution of World Bank Lending and Nonlending Services, 1975–95*  
(staff-weeks unless otherwise noted)

| <i>Period</i> | <i>Annual commitments<br/>(1987 dollars)</i> | <i>Total lending</i> |                    |                                                     | <i>Nonlending</i>    |                            |                                                     |
|---------------|----------------------------------------------|----------------------|--------------------|-----------------------------------------------------|----------------------|----------------------------|-----------------------------------------------------|
|               |                                              | <i>Preparation</i>   | <i>Supervision</i> | <i>Staff-weeks per<br/>\$1 million<br/>in loans</i> | <i>Macroeconomic</i> | <i>Sector-<br/>related</i> | <i>Staff-weeks per<br/>\$1 million<br/>in loans</i> |
| 1975–76       | 12.14                                        | 24,265               | 11,326             | 2.93                                                | —                    | —                          | —                                                   |
| 1977–80       | 14.28                                        | 29,267               | 16,443             | 3.20                                                | —                    | —                          | —                                                   |
| 1981–84       | 16.03                                        | 28,657               | 21,359             | 3.12                                                | —                    | —                          | —                                                   |
| 1985–88       | 17.32                                        | 31,819               | 22,783             | 3.15                                                | 6,917                | 8,604                      | 0.90                                                |
| 1989–92       | 19.09                                        | 36,039               | 25,172             | 3.21                                                | 7,152                | 11,819                     | 0.99                                                |
| 1993–95       | 18.26                                        | 36,702               | 34,621             | 3.91                                                | 9,310                | 17,225                     | 1.45                                                |
| Total         | 16.47                                        | 31,512               | 22,359             | 3.27                                                | 9,143                | 13,336                     | 1.36                                                |

— Not available.

Note: Values are annual averages.

Source: Authors' calculations based on World Bank data.

sion of spending on lending during the same period (0.46 compared with 0.7 staff-week per \$1 million in loans), implying that in relative terms the importance of ESW actually decreased slightly.

A breakdown by sector for 1985–90 and 1991–95 highlights that the relative constancy of real commitments in the aggregate hides important intersectoral variations (table 2). Regarding the volume of lending, considerably reduced levels of commitments in industry and energy and in agriculture (down \$0.94 billion and \$0.71 billion a year, respectively) were more than compensated by increases in lending for education, health, and public sector reform (up \$0.92 billion, \$0.67 billion, and \$0.54 billion, respectively). The ranking of sectors was little affected; the largest volume of lending was for infrastructure, followed by industry and energy and by agriculture and environment. The relative importance of education and health more than doubled.

In 1985–90, direct lending costs, in staff-weeks per \$1 million in loans, were above average in health (7.3), education (4.5), agriculture (4.4), and public sector reform (3.9). They were below average in infrastructure (2.9), industry (2.3), and financial sector operations (2.0). Turning to intertemporal changes, the cost of lending (normalized by volume of total lending) decreased in health and education. It increased considerably in agriculture, where, in contrast to industry and energy, the decrease in overall commitments was not matched by a concomitant reduction in staff-weeks. While ESW increased in every sector, the increase was, in quantitative terms, largest in infrastructure, where the amount of resources used for such services almost tripled. In 1991–95, the share of total sectoral resources (per \$1 million in loans) spent on ESW ranged from a high of 28 percent in industry and energy to a low of about 16 percent in health and agriculture (divide column 4 by [column 4 + column 2] in table 2).

Table 3 shows the regional disaggregation of lending and nonlending services. The largest increase in commitments was registered by Eastern and Central Europe followed by East Asia and the Pacific (up \$1.44 billion and \$0.96 billion per year, respectively). By contrast, annual real lending to South Asia dropped by almost \$1.2 billion, with smaller decreases in Latin America and Africa. The share of ESW in total use of staff time during 1991–95 was about 20 percent for almost all regions except Latin America (15 percent). This implies that staff time spent directly in both lending services and ESW varied widely across countries from a total of more than 10 staff-weeks per \$1 million in loans in Africa to less than 3 staff-weeks in East Asia. Comparatively high costs of lending were also observed in South Asia (a sharp increase from the 1985–90 period) and in the Middle East and North Africa.

### *Measures of Project Success*

Direct indicators regarding the success of individual lending projects are available from ex post evaluations conducted by the World Bank's Operations Evaluation Department (OED) and from the Annual Review of Portfolio Performance (ARPP). While the former are preferred insofar as they provide an assessment of

Table 2. *World Bank Resources Spent on Lending and Nonlending Services by Sector, 1985–90 and 1991–95*

| Sector                      | Annual commitments<br>in billions of<br>1987 dollars<br>(1) | Lending services   |                                                   | Nonlending services |                                                   |
|-----------------------------|-------------------------------------------------------------|--------------------|---------------------------------------------------|---------------------|---------------------------------------------------|
|                             |                                                             | Staff-weeks<br>(2) | Staff-weeks per<br>\$1 million<br>in loans<br>(3) | Staff-weeks<br>(4)  | Staff-weeks per<br>\$1 million<br>in loans<br>(5) |
| 1985-90                     |                                                             |                    |                                                   |                     |                                                   |
| Agriculture and environment | 3.81                                                        | 16,853             | 4.42                                              | 2,535               | 0.67                                              |
| Public sector reform        | 0.25                                                        | 972                | 3.89                                              | 255                 | 1.02                                              |
| Education                   | 0.92                                                        | 4,092              | 4.45                                              | 964                 | 1.05                                              |
| Health                      | 0.39                                                        | 2,815              | 7.28                                              | 664                 | 1.72                                              |
| Industry and energy         | 4.37                                                        | 10,230             | 2.34                                              | 2,828               | 0.65                                              |
| Infrastructure              | 4.22                                                        | 12,083             | 2.86                                              | 1,170               | 0.28                                              |
| Finance                     | 2.08                                                        | 4,052              | 1.95                                              | 1,045               | 0.50                                              |
| Total                       | 16.04                                                       | 51,097             | 3.19                                              | 9,460               | 0.59                                              |
| 1991-95                     |                                                             |                    |                                                   |                     |                                                   |
| Agriculture and environment | 3.10                                                        | 18,694             | 6.02                                              | 3,688               | 1.19                                              |
| Public sector reform        | 0.79                                                        | 3,282              | 4.14                                              | 696                 | 0.88                                              |
| Education                   | 1.84                                                        | 7,488              | 4.07                                              | 1,792               | 0.97                                              |
| Health                      | 1.06                                                        | 6,543              | 6.19                                              | 1,220               | 1.15                                              |
| Industry and energy         | 3.43                                                        | 9,576              | 2.79                                              | 3,728               | 1.09                                              |
| Infrastructure              | 4.39                                                        | 14,491             | 3.30                                              | 3,117               | 0.71                                              |
| Finance                     | 1.48                                                        | 3,675              | 2.48                                              | 1,080               | 0.73                                              |
| Total                       | 16.09                                                       | 63,749             | 3.96                                              | 15,320              | 0.95                                              |

*Source:* Authors' calculations based on World Bank data.

Table 3. *World Bank Resources Spent on Lending and Nonlending Services by Region, 1985–90 and 1991–95*

| Region                          | Annual commitments<br>(billions of<br>1987 dollars) | Lending services |                                            | Nonlending services |                                            |
|---------------------------------|-----------------------------------------------------|------------------|--------------------------------------------|---------------------|--------------------------------------------|
|                                 |                                                     | Staff-weeks      | Staff-weeks per<br>\$1 million<br>in loans | Staff-weeks         | Staff-weeks per<br>\$1 million<br>in loans |
| 1985-90                         |                                                     |                  |                                            |                     |                                            |
| Africa                          | 2.81                                                | 18,519           | 6.58                                       | 3,270               | 1.16                                       |
| East Asia and the Pacific       | 3.60                                                | 8,785            | 2.44                                       | 1,982               | 0.55                                       |
| Middle East and North Africa    | 1.25                                                | 5,103            | 4.09                                       | 880                 | 0.71                                       |
| Latin America and the Caribbean | 4.99                                                | 10,007           | 2.00                                       | 1,542               | 0.31                                       |
| Eastern and Central Europe      | 1.54                                                | 3,623            | 2.35                                       | 471                 | 0.31                                       |
| South Asia                      | 3.77                                                | 9,261            | 2.45                                       | 1,314               | 0.35                                       |
| Total                           | 17.97                                               | 55,298           | 3.08                                       | 9,460               | 0.53                                       |
| 1991-95                         |                                                     |                  |                                            |                     |                                            |
| Africa                          | 2.55                                                | 21,152           | 8.29                                       | 4,975               | 1.95                                       |
| East Asia and the Pacific       | 4.56                                                | 10,531           | 2.31                                       | 2,508               | 0.55                                       |
| Middle East and North Africa    | 1.25                                                | 4,864            | 3.88                                       | 1,260               | 1.01                                       |
| Latin America and the Caribbean | 4.60                                                | 12,322           | 2.68                                       | 2,322               | 0.51                                       |
| Eastern and Central Europe      | 2.98                                                | 9,026            | 3.03                                       | 1,983               | 0.67                                       |
| South Asia                      | 2.58                                                | 11,184           | 4.33                                       | 2,272               | 0.88                                       |
| Total                           | 18.52                                               | 69,079           | 3.73                                       | 15,320              | 0.83                                       |

Source: Authors' calculations based on World Bank data.

Table 4. *Indicators of the Success of World Bank Projects by Sector and Region, 1975–95*  
(mean values)

| <i>Sector or region</i>         | <i>OED indicator</i>                         |                                    |                                  |                               | <i>ARPP indicator</i>         |                                |
|---------------------------------|----------------------------------------------|------------------------------------|----------------------------------|-------------------------------|-------------------------------|--------------------------------|
|                                 | <i>Economic rate of return at completion</i> | <i>Project outcome<sup>a</sup></i> | <i>Institutional development</i> | <i>Project sustainability</i> | <i>Development objectives</i> | <i>Implementation progress</i> |
| <i>Sector</i>                   |                                              |                                    |                                  |                               |                               |                                |
| All                             | 16.81                                        | 71.71                              | 2.69                             | 2.40                          | 3.39                          | 3.10                           |
| Agriculture and environment     | 13.83                                        | 62.52                              | 2.44                             | 2.34                          | 3.25                          | 2.98                           |
| Public sector reform            | —                                            | 48.00                              | 2.38                             | 2.16                          | 3.21                          | 3.07                           |
| Education                       | —                                            | 82.33                              | 3.19                             | 2.53                          | 3.44                          | 3.17                           |
| Health                          | —                                            | 60.32                              | 2.59                             | 2.22                          | 3.25                          | 3.01                           |
| Industry and energy             | 15.45                                        | 74.69                              | 2.87                             | 2.50                          | 3.49                          | 3.16                           |
| Infrastructure                  | 20.88                                        | 77.97                              | 2.78                             | 2.38                          | 3.50                          | 3.14                           |
| Finance                         | 17.14                                        | 75.08                              | 2.76                             | 2.54                          | 3.42                          | 3.13                           |
| <i>Region</i>                   |                                              |                                    |                                  |                               |                               |                                |
| All                             | 16.81                                        | 71.71                              | 2.69                             | 2.40                          | 3.39                          | 3.10                           |
| Africa                          | 14.73                                        | 61.82                              | 2.32                             | 2.18                          | 3.25                          | 3.04                           |
| East Asia and the Pacific       | 18.95                                        | 82.83                              | 3.22                             | 2.79                          | 3.56                          | 3.30                           |
| Middle East and North Africa    | 17.82                                        | 77.61                              | 2.94                             | 2.57                          | 3.47                          | 3.10                           |
| Latin America and the Caribbean | 15.72                                        | 68.86                              | 2.67                             | 2.37                          | 3.40                          | 3.07                           |
| Eastern and Central Europe      | 16.95                                        | 75.49                              | 2.94                             | 2.60                          | 3.36                          | 3.04                           |
| South Asia                      | 18.97                                        | 76.98                              | 2.69                             | 2.29                          | 3.42                          | 3.07                           |
| Number of observations          | 1,959                                        | 3,956                              | 2,057                            | 1,992                         | 5,004                         | 5,064                          |

— Not available.

*Note:* OED indicators are from the World Bank's Operations Evaluation Department. ARPP indicators are from the World Bank's Annual Review of Portfolio Performance.

a. The share of projects that are rated "satisfactory."

*Source:* Authors' calculations based on World Bank data.

the overall performance of a completed project, the greater availability of data for ARPP indicators leads us to use them as an alternative variable to check the robustness of our results (although the corresponding regressions are not reported).

OED indicators, which are established by OED staff, include an estimate of the economic rate of return at completion, a binary measure (satisfactory and unsatisfactory) of the overall project outcome, an ordinal assessment (four levels from highly satisfactory to highly unsatisfactory) of the project's sustainability, and its contribution to institutional development in the client country. Although these indicators provide a succinct summary of overall project success, their disadvantage is that only completed projects undergo this evaluation. Because information on resources devoted to nonlending services is available on a consistent basis only after 1985, the number of projects for which both OED evaluations and data on nonlending services are available is substantially reduced. However, this drawback may be more than compensated for by the fact that the evaluation is undertaken by an independent unit, presumably with some minimum consistency across projects.

The ARPP rates projects annually according to several criteria relating to overall project performance, that is, contribution to development objectives and implementation progress. It also rates projects on more specific intermediate characteristics, such as compliance with legal covenants, project management performance, availability of funds, procurement progress, training progress, technical assistance progress, studies progress, environmental aspects, fiscal performance, and gender aspects. Not all of these criteria and characteristics are rated regularly, resulting in large variations in the number of observations available for different indicators. The analytical rigor and informational content of these measures are likely to be inferior to more precise indicators such as the economic rate of return. However, the much increased coverage partly compensates for this shortcoming, with a total of more than 5,000 projects having been included in the ARPP at least once. At the project level, the advantage of greater data availability is less impressive than might appear at first sight. Although using the mean value of all available ratings for any given project to construct a project-specific performance indicator increases the number of observations, it lumps together projects at very different stages of the project life cycle, whereas use of the final rating (for completed projects) considerably reduces the number of observations.

Differences in average values for selected indicators of project success are illustrated in table 4. Data on economic returns after completion are available for about 2,000 projects in four sectors (agriculture and environment, industry and energy, infrastructure, and finance). Average ex post rates of return vary between 13.8 percent in agriculture and 20.9 percent in infrastructure and between 14.7 percent in Africa and 19.0 percent in South and East Asia. Ex post ratings concerning contribution to institutional development and sustainability are highest in infrastructure and education projects. Contemporaneous (ARPP)

Table 5. *Simple Correlations between Different Indicators of Success of World Bank Projects, 1975–95*

| Sector or region                      | OED indicator                |                                       |                        |                           | ARPP indicator         |                         |
|---------------------------------------|------------------------------|---------------------------------------|------------------------|---------------------------|------------------------|-------------------------|
|                                       | Project outcome <sup>a</sup> | Economic rate of return at completion | Project sustainability | Institutional development | Development objectives | Implementation progress |
| <i>OED indicator</i>                  |                              |                                       |                        |                           |                        |                         |
| Project outcome <sup>a</sup>          | 1<br>[3,957]                 |                                       |                        |                           |                        |                         |
| Economic rate of return at completion | 0.3807***<br>[1,376]         | 1<br>[1,376]                          |                        |                           |                        |                         |
| Project sustainability                | 0.6381***<br>[2,024]         | 0.3107***<br>[842]                    | 1<br>[2,031]           |                           |                        |                         |
| Institutional development             | 0.4829***<br>[1,956]         | 0.1477***<br>[799]                    | 0.5084<br>[1,951]      | 1<br>[1,968]              |                        |                         |
| <i>ARPP indicator</i>                 |                              |                                       |                        |                           |                        |                         |
| Development objectives                | 0.2035***<br>[2,885]         | 0.1522***<br>[1,371]                  | 0.1791***<br>[2,024]   | 0.1355***<br>[1,960]      | 1<br>[4,806]           |                         |
| Implementation progress               | 0.2134***<br>[2,928]         | 0.1741***<br>[1,375]                  | 0.2447**<br>[2,028]    | 0.2184***<br>[1,965]      | 0.5719***<br>[4,805]   | 1<br>[4,855]            |

\*\* Significant at 5 percent.

\*\*\* Significant at 1 percent.

Note: OED indicators are from the World Bank's Operations Evaluation Department. ARPP indicators are from the World Bank's Annual Review of Portfolio Performance. Values in square brackets are the number of observations.

a. The share of projects that are rated "satisfactory."

Source: Authors' calculations based on World Bank data.



ratings, although characterized by limited variation across projects, are lowest in agriculture and in Africa and are consistently high in East Asia. Table 5 provides simple correlations between these indicators, illustrating that ARPP indicators for achievement of development objectives and implementation progress, OED indicators for project outcome, sustainability, and institutional development, as well as the ex post rate of return are significantly correlated with one another.

### *Policy Variables*

The literature has long emphasized that policies are important factors influencing project implementation as well as eventual development impact. It is therefore important to control for appropriate policy variables. Policies that have an important bearing on project success relate to fiscal policy, monetary policy, and external trade policy. These can be approximated by the public sector deficit, domestic inflation or the black market exchange rate, and openness, as in Burnside and Dollar (1996). To avoid problems of endogeneity, we use the value of policy variables at project initiation throughout. Note that policy variables are generally available only up to 1992, causing us to omit them in the instrumental equations for supervision.

### *Preliminary Statistical Analysis*

Investigation of the bivariate relationship between OED success indicators for the lending program and our independent variables (table 6) leads to three broad conclusions. First, the correlation between ESW (both macroeconomic and sector-specific) and development impact is significant and positive except in the case of project outcome. Second, higher levels of resources spent on supervision and preparation are negatively, rather than positively, related to project success. Such a phenomenon could arise from selection bias whereby bad projects are allocated more resources in order to improve their performance. Alternatively, resources spent in either preparation or supervision may be correlated with another omitted variable (it could be, for example, an Africa dummy) that is associated with lower project performance. And third, concerning the impact of policy variables, we find that openness is consistently associated with improved project success, that sustainability and contribution to institutional development are better in countries with low public sector deficits, that inflation negatively affects project outcome and sustainability, and that projects appear to perform better in all dimensions except economic rate of return in countries that invest a higher proportion of their own resources.

Table 7 uses descriptive statistics to summarize our hypotheses and to provide empirical support for the conjecture of a positive link between ESW and project success. Splitting the sample of completed projects into groups based on the overall project outcome, the variable used in our analysis, we find that the satisfactory group of projects benefited from considerably higher levels of ESW but required significantly lower levels of preparation and supervision (per \$1

Table 6. *Correlations between World Bank Program Level OED Indicators and Economic Sector Work as well as Policy Variables, 1975–95*

| Policy variable           | OED indicator                         |                              |                        |                           |
|---------------------------|---------------------------------------|------------------------------|------------------------|---------------------------|
|                           | Economic rate of return at completion | Project outcome <sup>a</sup> | Project sustainability | Institutional development |
| Analytical services (ESW) |                                       |                              |                        |                           |
| Total                     | 0.1538***<br>[305]                    | 0.0715<br>[347]              | 0.1331**<br>[601]      | 0.1237***<br>[595]        |
| Macro only                | 0.13368**<br>[305]                    | 0.05633<br>[347]             | 0.1269***<br>[601]     | 0.1102***<br>[595]        |
| Sectoral only             | 0.1525***<br>[305]                    | 0.07473<br>[347]             | 0.1432***<br>[601]     | 0.1215***<br>[595]        |
| Preparation               | -0.1139***<br>[1,299]                 | -0.0507*<br>[1,080]          | -0.0499*<br>[1,500]    | -0.0214<br>[1,474]        |
| Supervision               | -0.1218***<br>[1,299]                 | -0.0751**<br>[1,080]         | -0.0903***<br>[1,500]  | -0.0309<br>[1,474]        |
| Openness                  | 0.0495*<br>[1,225]                    | 0.0914***<br>[1,008]         | 0.1303***<br>[1,391]   | 0.0978***<br>[1,368]      |
| Public sector surplus     | 0.0361<br>[706]                       | 0.0488<br>[587]              | 0.0993***<br>[894]     | 0.0830**<br>[881]         |
| Inflation                 | -0.0438<br>[1,095]                    | -0.0955***<br>[927]          | -0.0575**<br>[1,287]   | -0.0241<br>[1,269]        |
| Investment                | -0.01121<br>[1,225]                   | 0.1174***<br>[1,008]         | 0.1999***<br>[1,391]   | 0.1554***<br>[1,368]      |

\* Significant at 10 percent.

\*\* Significant at 5 percent.

\*\*\* Significant at 1 percent.

Note: The lower number of observations is due to the use of portfolio-level data. Values in square brackets are the number of observations. Policy variables are from the Summers/Heston data set (openness and investment), the World Development Indicators (inflation), and William Easterly (public sector surplus and black market premium).

a. The share of projects that are rated "satisfactory."

Source: Authors' calculations based on World Bank data.

million in loans) than the rest. Key policies for the satisfactory projects were more favorable, with more open economies, lower public sector deficits, lower inflation and black market premia, and higher levels of investment than for the unsatisfactory group. However, the fact that unsatisfactory projects required 20 percent more preparation and 22 percent more supervision suggests that the ex ante design of a project, an activity to which ESW presumably contributes, is crucial. We obtained similar results (not reported) when we split projects into top and bottom groups for the other indicators discussed earlier. Although similar descriptive results hold if ARPP indicators are used, we refrain from using them because the only advantage of ARPP over OED ratings—namely, the inclusion of projects that are not yet completed—is outweighed by the difficulty of appropriately aggregating ratings and by the fact that they may be subjective and prone to changes due to shifts in personnel.

Table 7. *Input and Policy Indicators for Satisfactory and Unsatisfactory World Bank Projects, 1975–95*

| <i>Indicator</i>                                            | <i>Satisfactory outcome</i> | <i>Unsatisfactory outcome</i> |
|-------------------------------------------------------------|-----------------------------|-------------------------------|
| Economic sector work (staff-weeks)                          | 88.32                       | 74.33                         |
| Preparation of loans (staff-weeks per \$1 million in loans) | 4.88                        | 5.84                          |
| Supervision of loans (staff-weeks per \$1 million in loans) | 4.75                        | 5.82                          |
| Openness                                                    | 54.11                       | 48.53                         |
| Public sector surplus                                       | –6.09                       | –6.57                         |
| Inflation                                                   | 23.56                       | 49.69                         |
| Black market premium                                        | 28.37                       | 35.16                         |
| Investment                                                  | 14.97                       | 12.71                         |

*Note:* Policy variables are from the Summers/Heston data set (openness and investment), the World Development Indicators (inflation), and William Easterly (public sector surplus and black market premium).

*Source:* Authors' calculations based on World Bank data.

#### IV. EMPIRICAL ANALYSIS OF THE IMPACT OF ESW

In this section, we report the results of the reduced-form regressions discussed in section II. We examine whether, and if so by how much, ESW improves project quality. We evaluate whether the allocation of staff resources across three main uses—ESW, preparation, and supervision—has been consistent with the maximization of development impact. And we analyze whether the allocation of staff resources to ESW reduces the volume of lending, presumably by reducing the amount of staff time available for the more immediate tasks of project processing.

##### *Estimating the Country Manager Model*

We first estimate the reduced-form equation 1. Because we are interested in the overall program outcome, we use the mean satisfactory/unsatisfactory rating for all projects (weighted by loan size) as the dependent variable. (The results do not change much if the unweighted average outcome of the lending program is used.) Where included, investment lending is instrumented using past preparation and supervision requirements in the country, past project success, and the mean project age in a given country and year.

The results in the first column in table 8 suggest that ESW has a significant positive impact on the quality of the lending program, thus supporting hypothesis 1. This effect remains significant if policy variables are added to the regression. Because our measure for ESW includes the years before project approval while policies here are measured at project approval, there is a potential endogeneity problem. Repeating the analysis with policies in earlier years reduces the number of observations but does not alter the substantive conclusions (results not reported).

Thus the use of better data provides empirical evidence in favor of a significant and positive impact of ESW on quality, in contrast with the findings of Arias (1994). Regarding the quantitative impact of ESW, we find that an additional 100 staff-weeks would increase the probability of a program being satisfactory

Table 8. *Determinants of Project Outcome and the Economic Rate of Return for World Bank Loans, 1975–95*

| Variable                        | Project outcome <sup>a</sup> |                      |                        | Economic rate of return |                     |                      |
|---------------------------------|------------------------------|----------------------|------------------------|-------------------------|---------------------|----------------------|
|                                 | 1                            | 2                    | 3                      | 4                       | 5                   | 6                    |
| ESW                             | 0.12104**<br>(0.0529)        | 0.1760**<br>(0.0753) | 0.1773*<br>(0.1008)    | 1.933***<br>-0.7132     | 2.240**<br>(0.9319) | 2.1605*<br>(1.243)   |
| Openness                        |                              | 0.36271<br>(0.3394)  | 0.3778<br>(0.4178)     |                         | -2.262<br>(4.647)   | 0.6309<br>(4.778)    |
| Inflation                       |                              | -0.2451<br>(0.1403)  | -0.3372**<br>(0.01717) |                         | -0.0778<br>(0.5883) | -0.7225<br>(0.5187)  |
| Public sector surplus           |                              | 0.8937<br>(1.538)    | 0.8037<br>(1.707)      |                         | -2.917<br>(2.964)   | -37.215<br>(0.31215) |
| Investment lending <sup>b</sup> |                              |                      | 0.0054<br>(0.0221)     |                         |                     | 0.0709<br>(0.2227)   |
| Log-likelihood/R <sup>2</sup>   | -224.99                      | -136.52              | -98.96                 | 0.02                    | 0.05                | 0.12                 |
| Number of observations          | 347                          | 215                  | 157                    | 305                     | 196                 | 164                  |

\* Significant at 10 percent.

\*\* Significant at 5 percent.

\*\*\* Significant at 1 percent.

Note: Policy variables are from the Summers/Heston data set (openness and investment), the World Development Indicators (inflation), and William Easterly (public sector surplus and black market premium). Standard errors are in parentheses.

a. The mean satisfactory/unsatisfactory rating for all projects, weighted by loan size.

b. Staff-weeks spent on supervision and preparation for investment lending.

Source: Authors' calculations.

by between 12 and 20 percent. While other resources spent on lending—supervision and preparation—are insignificant, the point estimate for ESW is positive (the third column in table 8).<sup>3</sup> And although the policy variables have the anticipated sign (and can be shown to be significant in regressions when ESW is excluded), they are not significant. Using other OED variables to represent project quality suggests that the results obtained are robust (not reported). While policy variables are rarely significant (only public sector surplus in the sustainability equation), ESW is highly significant throughout.

The regressions using the economic rate of return (ERR) as the dependent variable have the added advantage of facilitating a simple cost-benefit analysis of assigning more resources to ESW. However, several problems arise with using ERRs. ERRs are only available for certain sectors (see table 4), and even within these sectors ERRs have not been computed consistently for all projects. This implies that whatever estimate is available is unlikely to be representative for the entire lending program, and problems of sample selection bias may be present. Also, aggregation of ERRs over projects (even if the latter are weighted by loan size) is at best a rough approximation of the true net present value (NPV) of the lending program. And the number of observations is quite limited (305 and 164 without and with policy variables, respectively).

3. The point estimate for INV is negative, though insignificant, in a specification that excludes policies (not reported). This suggests that, with the instruments available, we were not able to resolve fully the endogeneity problem associated with project preparation and supervision.

These problems prevent us from reading too much into the figures obtained in the respective regressions. However, ESW is significant and positive even if policy variables are included (table 8). Introducing regional dummies (not reported) does not eliminate the significance of the coefficient on ESW. In fact, using the point estimate on the ESW variable in the regressions, we find that an increase of one staff-week in ESW would increase the NPV of the average lending program (\$210 million) by between \$36,000 and \$44,000, similar to the higher end of the range of values (\$18,000–\$36,000) derived from the project-level regression reported here.<sup>4</sup> Compared with the average cost of one staff-week (about \$3,000) as reported in World Bank data, this suggests that an expansion in ESW would constitute an economically sound investment.

When interpreting this result, it is important to keep in mind the distinction between an improvement in the NPV for projects supported by the World Bank and an improvement in the NPV for the government's investment program. If ESW enables World Bank staff to identify and support new investment possibilities, thereby expanding the range of investment options open to a country, or if it leads to an improvement in the *ex ante* design of a project already in the investment program and hence its ultimate development impact, then ESW yields a genuine return to the recipient country. Alternatively, conducting more ESW may enable World Bank staff to cherry-pick better projects from the existing portfolio, projects that would have been undertaken by the country anyway. In this event, ESW improves the performance of the World Bank's lending portfolio but not necessarily the quality of the public investment portfolio in general. However, this explanation is not convincing. In most cases, countries are not in the position of having a well-defined investment program from which World Bank staff can pick and choose. World Bank staff are usually heavily engaged in identifying and developing new investment opportunities. And in most cases, projects are not close to being fully designed before World Bank staff become involved. In fact, World Bank staff usually play a major role in project design and conceptualization.

A more formal test of these alternative explanations is possible. If the cherry-picking explanation underlies the observed impact of ESW on the NPV of the World Bank-supported lending program, it would be expected that World Bank portfolios supported by significant inputs of ESW would comprise exclusively "safe" projects. Cherry-picking implies that few, if any, projects would be dropped either during preparation or after approval. Thus, the cherry-picking hypothesis leads us to expect an inverse relationship between the amount of ESW and the

4. The transformation from ERR to NPV depends on the phasing of costs and benefits. Following Kilby (1995), the calculation reported in the text assumes that project costs occur in the first period and benefits in the second. With this assumption and an average lending program of \$210 million, the increase in NPV lies between  $(\$0.00019 \times 210,000,000) / 1.1$  and  $(\$0.00023 \times 210,000,000) / 1.1$ , where the discount rate is assumed to be 10 percent. A more realistic assumption regarding the phasing of costs and benefits would tend to increase the calculated NPV. We therefore treat the value reported in the text as a conservative estimate.

share of the program dropped. No such relationship is found in the data. Controlling for policies and the level of gross domestic product per capita, there is no statistically significant relationship between the share of the lending program dropped prior to the loan becoming effective and the amount of ESW, nor between the share of the portfolio canceled after approval and the amount of ESW (results not reported). This result suggests that, rather than enabling task managers to pick the cherries out of a limited portfolio of investments, ESW contributes by either expanding the available investment opportunities or improving the ex ante design of the available projects. Both avenues result in a genuine increase in the development impact of foreign assistance.

Regarding a possible trade-off between ESW and investment lending (INV) as stated in hypothesis 2, we find that, contrary to expectation, the marginal contribution of ESW exceeds that of INV. Indeed, the marginal contribution of INV, while positive, is not significantly different from 0 (the third and sixth columns in table 8). This suggests that it would have been possible to shift resources from INV to ESW and increase program quality. And because ESW also (we assume) contributes to policy impact, we conclude that the observed allocation of staff time between these two uses has not been optimal from the point of view of maximizing development impact. In other words, our ideal model of decision-making at the level of the country manager may not be a good approximation of reality.

Of course, the representative manager may not have been concerned exclusively with development impact as we have assumed. Other objectives are clearly possible. For example, managers may have given weight to resource flows or lending volume. This could reflect a concern with the overall level of financial transfer (a development concern) or, if the volume of commitments is the basis for rewards within the institution, a career concern. Claessens, Pohl, and Quian (1996) find a significant impact of task managers on project quality but note that the quality of task managers is not correlated with promotion decisions within the World Bank. Either way, the objective function outlined in section II needs to be expanded to include a third element—lending volume—in addition to the quality of lending and policy impact. Corresponding to equation 1, we now have an additional reduced-form equation for the optimal lending volume  $V_{jt}^*$  consistent with our revised objective function for the country manager. Again assuming a linear approximation, the new equation to be estimated is:

$$(5) \quad V_{jt}^* = \lambda_0 + \lambda_1 \text{ESW}_{jt} + \lambda_2 \text{INV}_{jt} + \lambda_3 \text{POL}_{jt} + \lambda_4 \text{ADJ}_{jt} + v_{jt}$$

where, as before, ESW represents time spent on economic and sector work, INV is staff time devoted to lending services excluding adjustment operations, POL represents policy variables, and ADJ is the staff time for adjustment operations. Focusing on the difference between  $\lambda_2$  and  $\lambda_1$ , we can examine whether a reallocation of staff resources in favor of ESW would have a negative impact on lending volume (hypothesis 3). If  $\lambda_2 < \lambda_1$ , managers would have been operating within the frontier—that is, they could have increased both quality and volume of the

lending program by shifting resources out of investment and into ESW. If  $\lambda_2 \geq \lambda_1$ , reallocating staff from lending services to ESW, regardless of its impact on quality, would have come at the cost of decreased lending volume. This would imply a trade-off between quality and quantity, indicating that managers did indeed place some value on the volume of lending as well as the quality.

We use as a dependent variable both the volume of commitments and the volume of actual disbursements for a given lending program. The former is the better variable if the representative manager believed that approval of a large lending program by the World Bank's Board of Executive Directors was a key factor in promotions and career development. The latter is more appropriate if the manager was interested in achieving a large transfer of resources.

Our results (the first column in table 9) support the hypothesis of a significant difference between the effect of ESW and INV on the volume of commitments. INV appears to be between 40 and 50 percent more effective in increasing commitments than is ESW. If the observed relationship is indeed the outcome of an optimization process on the part of the decisionmaker, we can assess quantitatively the trade-off between volume and quality. Ignoring for the moment the possibility that ESW has a positive policy impact, at the optimum the decisionmaker will be indifferent to a marginal shift of staff time between INV and ESW. Using the estimated results for equations 1 and 5, we calculate that the average manager seems to be indifferent between a 1 percent decrease in the program's ERR (or a \$2 million decrease in the program's NPV) and an additional \$4 million in commitments (compared with an average lending program of \$210 million). This result suggests the presence of a strong trade-off. That is, World Bank managers are willing to accept a moderately large decline in the quality of the lending program in return for only a modest increase in the level of commitments. Of course, when making a decision about this trade-off, the country manager should

Table 9. *Determinants of World Bank Lending Volume, 1975-95*

| Variable                        | Commitments           | Disbursements          |
|---------------------------------|-----------------------|------------------------|
| ESW                             | 0.1899***<br>(0.039)  | 0.2118***<br>(0.04465) |
| Investment lending <sup>a</sup> | 0.2709***<br>(0.029)  | 0.0727**<br>(0.0314)   |
| Adjustment lending <sup>a</sup> | 0.1401***<br>(0.0338) | 0.1432***<br>(0.0324)  |
| Adjusted R <sup>2</sup>         | 0.9090                | 0.7248                 |
| Number of observations          | 406                   | 302                    |

\*\* Significant at 5 percent.

\*\*\* Significant at 1 percent.

Note: The dependent variable, lending volume, is commitments or disbursements in millions of dollars. Coefficients of policy variables and country dummies are included but not reported. Standard errors are in parentheses.

a. Staff-weeks spent on preparation and supervision.

Source: Authors' calculations.

also take account of the contribution of ESW to policy, an impact that is additional to its immediate effect on the quality of the lending program. If it is true that the contribution of ESW to policy impact is quantitatively important, then our unfavorable interpretation of managerial behavior is strengthened.

The picture with respect to disbursements is very different. In table 9 the evidence suggests that managers were operating within their frontier. That is, resources could have been switched from INV to ESW to the benefit of program quality and the volume of disbursements. In this case, managers could have achieved a 1 percentage point increase in the program's ERR (an increase in the NPV of about \$2 million) and increased disbursements by almost \$7 million. The different outcomes are consistent with the view that managers did give weight to the level of commitments, even though this reduced the quality of the lending program and reduced the level of disbursements.

### *Estimating the Task Manager Model*

We turn now to the decisionmaking problem faced by the task manager of a single project. In this case ESW is treated as an exogenous variable, and the manager is assumed to maximize project quality given the (shadow) price of staff inputs.

**PROJECT PREPARATION.** As discussed in section II, the assumed decisionmaking process yields demand functions for preparation and supervision contingent on existing ESW. According to the model, a higher level of ESW is expected to reduce the need for preparation resources because, as a result of ESW, the basic project rationale and design will be on a much firmer footing. Using total staff time spent on preparation (normalized by loan size) as the dependent variable and estimating equation 2 confirms this notion. In particular, we find that the representative manager clearly and significantly reduces the amount of resources used to prepare any given project as ESW increases (table 10). This result is consistent with the view that managers see a benefit from ESW and adjust their resource allocations accordingly. Regarding the magnitude of this impact, we find that, at the mean level of ESW across all projects (136 staff-weeks), one additional staff-week of ESW causes the task manager to reduce the time allocated to preparation by 0.015 staff-week per \$1 million in loans or 1.05 staff-weeks for the mean \$70 million loan (the third column in table 10). We also note that at empirically observed levels of input, the impact of ESW on preparation is concave, that is, it is subject to slowly decreasing returns (the first three columns in table 10). Returns from ESW start declining at almost 1,000 staff-weeks on average per project, a value that is well beyond the mean (136 staff-weeks). Because we are dealing with individual projects, we add dummies to control for unobservable sector-specific or country-specific characteristics. Their inclusion leaves the statistical results unchanged.

**PROJECT SUPERVISION.** According to our decisionmaking model, by improving the design of projects, ESW should enable managers to reduce the amount of time



Table 10. *Determinants of the Preparation Requirements for World Bank Projects, 1975–95*

| Variable                                    | 1                       | 2                     | 3                       | 4                     |
|---------------------------------------------|-------------------------|-----------------------|-------------------------|-----------------------|
| ESW                                         | -1.1784***<br>(0.143)   | -1.1907***<br>(0.222) | -2.0273***<br>(0.171)   | -0.2847***<br>(0.076) |
| ESW <sup>2</sup>                            | 0.0011***<br>(0.000)    | 0.0012***<br>(0.000)  | 0.0019***<br>(0.000)    |                       |
| Public sector surplus                       |                         | -5.2725*<br>(2.876)   |                         |                       |
| Inflation                                   |                         | -0.0408<br>(0.040)    |                         |                       |
| Openness                                    |                         | 0.9206*<br>(0.485)    |                         |                       |
| Country performance<br>(relative to region) |                         |                       |                         | 65.5442<br>(51.208)   |
| Past project success <sup>a</sup>           | -40.8768***<br>(15.185) | -39.9245*<br>(22.790) | -60.7319***<br>(14.018) | -42.7433<br>(32.492)  |
| Past preparation <sup>a</sup>               | 55.9327***<br>(3.753)   | 52.0307***<br>(5.789) |                         | 67.2006***<br>(6.239) |
| Adjusted R <sup>2</sup>                     | 0.1984                  | 0.1606                | 0.1115                  | 0.2224                |
| Number of observations                      | 2,355                   | 1,239                 | 3,194                   | 737                   |

\* Significant at 10 percent.

\*\*\* Significant at 1 percent.

Note: The dependent variable is total staff time spent on preparation (normalized by loan size). Sector dummies are included but not reported. Standard errors are in parentheses. Coefficients and standard errors are multiplied by 100.

a. At the country level.

Source: Authors' calculations.

allocated to supervision and thus have a negative effect on resource requirements for supervision (equation 3). Table 11 provides partial support that this is indeed the case; higher levels of ESW are clearly associated with lower inputs into supervision except when preparation is included as an explanatory variable. The other regressions reported in table 11 support the notion that, whether consciously or not, task managers recognize the benefits from ESW and, in line with the proposed maximization model, adjust the allocation of staff time to supervision accordingly. The quantitative magnitude of the effect of ESW on supervision is lower than in the case of preparation; an increase of 1 staff-week in ESW is associated with a decrease of 0.009 staff-week in supervision per \$1 million in loans. The decrease is equivalent to a reduction in supervision requirements by 0.64 staff-week for the mean \$70 million project (the first column in table 11). Furthermore, it is much cheaper to implement projects in countries with an appropriate policy environment. A 3 percentage point decrease in the public sector deficit, for example, reduces the amount of supervision needed by 0.27 staff-week per \$1 million in loans or 56.7 staff-weeks for the mean lending program of \$210 million (the second column in table 11).

Table 11. *Determinants of the Supervision Requirements for World Bank Projects, 1975–95*

| Variable                          | 1                       | 2                      | 3                       |
|-----------------------------------|-------------------------|------------------------|-------------------------|
| ESW                               | -1.5812***<br>(0.300)   | -1.0446***<br>(0.320)  | -0.2464<br>(0.250)      |
| ESW <sup>2</sup>                  | 0.0024***<br>(0.001)    | 0.0014**<br>(0.001)    | 0.0004<br>(0.001)       |
| Public sector surplus             |                         | -9.0527***<br>(2.872)  | -7.6614***<br>(2.222)   |
| Inflation                         |                         | 0.0769<br>(0.062)      | 0.0781<br>(0.048)       |
| Openness                          |                         | -1.2403**<br>(0.543)   | -0.8083*<br>(0.420)     |
| Loan size                         | -2.0057***<br>(0.145)   | -1.8145***<br>(0.151)  | -0.7596***<br>(0.123)   |
| Preparation                       |                         |                        | 63.4220***<br>(2.410)   |
| Past project success <sup>a</sup> | -73.1693***<br>(19.044) | -60.5990**<br>(23.937) | -57.4011***<br>(18.519) |
| Adjusted R <sup>2</sup>           | 0.2134                  | 0.228                  | 0.538                   |
| Number of observations            | 1,639                   | 1,045                  | 1,045                   |

\* Significant at 10 percent.

\*\* Significant at 5 percent.

\*\*\* Significant at 1 percent.

Note: The dependent variable is total staff time spent on supervision (normalized by loan size). Sector dummies are included but not reported. Standard errors are in parentheses.

a. At the country level.

Source: Authors' calculations.

Taken together with the results for project preparation, this implies that assigning one more staff-week to ESW would lead task managers to reduce the time allocated to project preparation and supervision per project by about 1.7 staff-weeks over the entire life of the project. Moreover, ESW actually benefits more than one project. If we assume that macroeconomic work benefits all projects in the lending program (3 on average for each country), and that sectoral analysis only benefits projects in a particular sector, we calculate that each staff-week of ESW benefits 1.5 projects on average. For the average investment project, the ratio of sector-specific to macroeconomic ESW is about 3 to 1. Thus the saving in preparation and supervision for the entire lending program generated by 1 staff-week of ESW amounts to 2.5 staff-weeks. Therefore, a focus on volume of lending that comes at the cost of project quality (as evidenced at the country level) does seem to increase the costs associated with administering World Bank loans.

**PROJECT QUALITY.** This evidence implies that task managers recognize the positive impact of ESW on lending quality. This hypothesis can be tested directly by estimating equation 4 at the project level for a wide range of independent variables and specifications. The variables of greatest interest are the project

outcome rating (satisfactory-unsatisfactory) and the ERR (estimated after project completion). We also use ARPP ratings (either the average over the whole project life cycle or the final rating for a completed project) to construct a large number of quality indicators that we then use as dependent variables. Because the results from these regressions do not add any substantively new insights, we do not report them separately.

From the probit equation of project outcome, ESW does significantly increase the probability of a satisfactory project outcome, even if other policy variables are included (the first three columns in table 12). An increase of 100 staff-weeks allocated to ESW increases the probability of a successful project by between 9 and 13 percent, comparable to the estimate obtained at the program level. In line with the literature, we find that better policies—lower levels of inflation and a more open economy—also contribute significantly to greater project success and are quantitatively quite important.

ERRs are not available for projects in the social sectors. And a project's benefit to the borrowing country is determined by the size of the loan as well as the ERR, so the net present value would be a more appropriate measure of the benefit to the borrower country. Unfortunately, the cost and benefit streams underlying the ERR calculation are available only for very few projects, so we cannot reproduce NPVs. That said, our results suggest that ESW causes a considerable increase in the ERR for a typical project (columns 4–6 in table 12), regardless of whether the specification includes policy variables, instrumented preparation and supervision, or country or sector dummies (hypothesis 1). An increase of one staff-week in ESW before project initiation increases a project's rate of return by between 0.02 and 0.04 percentage point.

For the mean loan of \$70 million in the sample of completed projects, this result translates into an increase in NPV of between \$12,000 and \$25,000.<sup>5</sup> And because each staff-week of ESW includes macroeconomic and sectoral analysis, it should, as noted, benefit more than one project in the lending program. Using the earlier calculation that 1 staff-week of ESW benefits 1.5 projects implies an increase in NPV for each additional staff-week of ESW of between \$18,000 and \$36,000. This result is very similar to the one obtained when the lending program was the unit of analysis. Based on the reported cost of a staff-week, slightly below \$3,000, if the sample of projects available here is any guide, putting additional resources into ESW would be an economically attractive option.<sup>6</sup> The fact that ESW is similarly significant in a probit regression of project outcomes suggests that this result is not driven by the sectoral limitations of the availability of ERRs.

5. Again following Kilby (1995), with an average project cost of \$70 million, a staff-week increase in ESW before project initiation increases net present value of an average project by between  $(\$0.0002 \times 70,000,000) / 1.1$  and  $(\$0.0004 \times 70,000,000) / 1.1$ , where the discount rate is assumed to be 10 percent.

6. Although the cost of a staff-week should include travel costs, the latter may not have been reported consistently in our data sources. Even a generous adjustment for this does not change the substantive results.

Table 12. *Probit Analysis of the Determinants of Project Outcome and the Economic Rate of Return for World Bank Loans, 1975–95*

| Variable                      | Project outcome <sup>a</sup> |                      |                       | Economic rate of return |                       |                      |
|-------------------------------|------------------------------|----------------------|-----------------------|-------------------------|-----------------------|----------------------|
|                               | 1                            | 2                    | 3                     | 4                       | 5                     | 6                    |
| ESW                           | 0.0973**<br>(0.387)          | 0.0900**<br>(0.407)  | 0.1302**<br>(0.063)   | 2.3737*<br>(1.4047)     | 4.2285***<br>(1.6638) | 2.3439*<br>(1.3990)  |
| Public sector surplus         |                              | 0.7192<br>(0.9632)   | 0.8727<br>(1.365)     |                         | 14.9735<br>(53.6540)  | 42.0955<br>(43.8989) |
| Inflation                     |                              | −0.0371*<br>(.0214)  | −0.06002**<br>(.0214) |                         | 1.9165<br>(10.2482)   | 4.90144<br>(5.798)   |
| Openness                      |                              | 0.7391***<br>(0.018) | 0.4729*<br>(0.2564)   |                         | −3.4949<br>(21.8876)  | −1.0494<br>(5.241)   |
| Preparation (instrumented)    |                              |                      | −0.5611*<br>(0.3192)  |                         |                       | 42.0955<br>(139.257) |
| Supervision (instrumented)    |                              |                      | 0.02102<br>(0.2649)   |                         |                       | 5.1640<br>(103.772)  |
| Log-likelihood/R <sup>2</sup> | −825.56                      | −534.62              | −394.05               | 0.146                   | 0.142                 | 0.1109               |
| Number of projects            | 1,324                        | 873                  | 643                   | 481                     | 302                   | 200                  |

\* Significant at 10 percent.

\*\* Significant at 5 percent.

\*\*\* Significant at 1 percent.

Note: Sector dummies included but not reported. Standard errors are in parentheses. Coefficients and standard errors are multiplied by 100.

a. The mean satisfactory/unsatisfactory rating for all projects, weighted by loan size.

Source: Authors' calculations.

Because equation 4 captures our central result (that lending quality depends on ESW), we subjected it to three tests of robustness. In the first, we changed the specification from linear to log-linear. With or without policy variables, the coefficient on ESW in the log-linear specification with either project outcome rating or ERR as the dependent variable was positive and significant at least at the 5 percent level (results not reported). In the second, we took advantage of new information that became available late in our study. The results reported thus far cover projects that were evaluated before July 1996. Having completed the estimation for projects up to this date, information became available on additional projects evaluated up to February 1998. Armed with this additional information, we reran equation 4. The results (not reported) are even stronger. They show that with or without policy variables, the coefficient on ESW remains positive and highly significant in the equation with project outcome rating as the dependent variable. In the equation with ERR as the dependent variable, the coefficient is positive and significant at the 5 percent level without policies and at the 10 percent level with policies.

The third test entails expanding the right-hand-side variables to include some others that have been found in the literature to have an impact on project outcomes. These include indexes of country risk, institutional quality, and bureaucratic delays (Knack and Keefer 1995) and the number of revolts, coups, and assassinations to capture political stability (King and Levine 1993). The results (not reported) for both the program-level and project-level regressions reveal that the coefficient on ESW never changes sign and remains significant in the majority of cases at the 10 percent level. Thus the results are robust to these tests of specification, sample selection, and choice of independent variables.

Although the task manager does not control the time allocated to ESW, we can still use equation 4 to test whether the country manager achieved the right allocation between lending services and ESW by exploring whether a shift in resources from preparation and supervision to ESW would have resulted in a better project outcome (hypothesis 2). Table 12 (the third and sixth columns) indicates that the marginal contribution of resources spent on supervision (instrumented) is positive but not statistically significant. The contribution of resources spent on preparation is also positive but insignificant in the equation for the rate of return and is negative and marginally significant in the equation for project outcome. We interpret this last result as an indication that, with the instruments available, we were not able to control for the endogeneity of these two variables, and not as an indication of a negative impact of preparation on project success. The general result is consistent with our results at the program level that an increase in the allocation of resources to ESW would have been associated with an overall improvement in project quality and presumably with a positive impact on policy.

As we saw earlier, there is a negative raw correlation between project quality and the amount of resources used in supervision (or preparation). We also note that supervision requirements are highly correlated with resource use in prepa-

ration (preparation alone explains about 55 percent of the variation in allocations to supervision). This correlation suggests that managers do not behave as though preparation and supervision are substitutes, as we have assumed. To the contrary, projects that consume substantial amounts of resources during preparation are generally associated with more than proportional allocations to supervision, possibly reflecting a different decisionmaking model.

Instead of a simultaneous decisionmaking process undertaken by a single task manager, we may be observing a sequential process undertaken by two or more managers. In the first step, staff time is allocated to preparation, given the availability of ESW to achieve maximum project quality at the time the project is approved by the board. This step reflects the fact that the project and its quality are subject to considerable scrutiny at this time and that staff are rewarded for performing well. In the second step, supervision resources are allocated, often by a different task manager, based on the project's quality at approval to maximize project quality by the time project implementation is completed. In this variant, the task manager who is preparing an ill-conceived project is obliged to allocate more time to preparation than would be the case for a project with a better conceptual foundation. But the task manager is not willing to allocate enough time to reduce future supervision requirements because someone else will be responsible for implementation. Hence, both preparation and supervision tend to be larger than normal for such projects. This suggests that the reduction in preparation that occurs when more ESW has been undertaken will carry over to a reduction in supervision as well.

There are certainly cases where higher spending on preparation (and supervision) may be justified, for example, in the development and piloting of untested or relatively complex approaches. But in general we interpret high preparation costs as indication of a failure to have carried out the necessary analytical background work or to have engaged in a constructive process of consensus building with the main stakeholders involved and, as a consequence, a lack of government commitment. Experience as well as our results suggest that in such situations it might be advisable first to prepare the ground with ESW before moving ahead with a lengthy and prolonged process of project preparation. Or, where this has not been done, the expected high costs of supervision should be borne in mind when decisions are being made on whether to drop or to continue preparing projects that have incurred large costs during preparation.

The fact that, even if instrumented, project preparation and supervision are never significantly different from 0 at the 5 percent level of significance suggests that, at the program as well as the project level, higher spending on preparation or supervision does not automatically increase quality. This failure to identify a positive contribution of preparation and supervision to project success does not mean that, at any given point in a project's life, higher spending on supervision will not increase the probability of an improvement in project quality as demonstrated by Kilby (1995). The difference is that Kilby looks at changes between successive ARPP ratings, thus eliminating project-specific fixed effects, while we

focus on the quality of the project as a whole. We therefore conclude that greater attention to ESW in the form of elaborating the broader context, clarifying the rationale for a specific type of intervention, and assessing its feasibility and economic desirability against a broader set of potential alternatives could result in the design of better projects and thus less need for spending time on preparation or supervision.

## V. CONCLUSION

At the start of this article, we posed three questions. In these concluding remarks, we report our answers to each one. We also try to explain why we get the results that we do regarding the behavior of managers. And we make suggestions for improving the impact that World Bank projects have on development.

First, does ESW improve the quality of World Bank loans? We find that ESW has a significant positive impact on various measures of the quality of World Bank projects. For example, an increase of one staff-week in the amount of time devoted to ESW before project initiation is associated with an increase in the economic rate of return for an individual project of between 0.02 and 0.04 percentage point. That translates into an increase of between \$12,000 and \$25,000 in the project's net present value for a cost of no more than \$3,000; \$1 of ESW yields \$4 to \$8 in development impact. Indeed, because each staff-week of ESW benefits more than one project, this is an underestimate. Examining the impact of ESW for the entire lending program for a country, we find that \$1 of ESW yields between \$12 and \$15 of development impact. Even this is an underestimate because our calculation fails to capture any benefit that ESW may have in terms of influencing policy formulation in developing countries, a key objective of ESW. We therefore conclude that there has been a high payoff to ESW.

Second, has the World Bank underinvested in ESW relative to lending services? Our results at both the program and the project level provide clear evidence of underinvestment in ESW. Assuming that staff resources are fungible across time and among different uses, the marginal contribution of ESW to the quality of lending should be lower than that of lending services (preparation and supervision), because ESW has benefits beyond its immediate impact on lending whereas lending services do not. Indeed, ESW is often undertaken to provide the basis for policy advice to governments and is not necessarily tightly linked to a particular project or lending program. However, we find the reverse, that ESW has a systematically positive effect on the quality of the lending program, whereas neither preparation nor supervision turns out to be significant. This result suggests that reallocating staff time from lending services to ESW would increase the quality of the lending program. Consistent with this, we find that task managers at the project level are able to reduce the time allocated to lending services by about 2.5 staff-weeks for every staff-week expended on ESW. We infer that ESW helps staff to identify and support new investment options and expands the set of feasible projects. It also helps staff to design better projects *ex ante* and im-

proves the quality of projects already in the investment program. Preparation and supervision, by contrast, can only improve the quality of a project (whether good or bad) *ex post*.

Third, is there a trade-off between the quality and volume of lending? We find that time allocated to preparation and supervision is between 40 and 50 percent more effective in increasing commitments than ESW, but that resources could be switched from preparation and supervision to ESW to the benefit of both the quality of program and level of disbursements. Even though the results suggest that higher levels of ESW improve the quality of the lending program, it is still possible that, within a given resource envelope, shifting staff time from lending services to ESW reduces the overall volume of lending. Addressing this question, we find that this is indeed the case. Lending services are between 40 and 50 percent more effective in increasing total commitments than is ESW. However, when we analyze disbursements (resource transfer) instead of commitments, we find that managers could increase both the quality of lending and disbursements by switching resources from lending services to ESW. This result, together with our conclusion that there has been underinvestment in ESW from the standpoint of project quality, suggests that at least to some degree the volume of commitments has been an additional objective guiding the disposition of staff resources. Our results provide some insight into the trade-off between quality and quantity. The analysis suggests that on average a manager is indifferent between a decrease of \$2 million in the NPV of a lending program and an additional \$4 million of lending volume. If this estimate is broadly accurate, it suggests that managers are prepared to allow a substantial reduction in program quality in return for only a small increase in commitments relative to the average program size.

Identifying the behavioral relationships underlying any set of data is a difficult task. Our analysis finds evidence that broad allocations of World Bank staff resources across different activities are influenced by a concern about commitments. Country managers choose to allocate resources in favor of preparation at the expense of ESW. And we find evidence that task managers clearly perceive the benefit of ESW and respond by reducing the amount of time allocated to preparation and supervision. Our results suggest that managers reduce preparation and supervision by about 2.5 staff-weeks if one more week is allocated to ESW. Together, these results are consistent with the view that task managers factor availability of ESW into their decisions regarding the allocation of resources at the project level at least to some extent, but that country managers systematically underinvest in ESW at the program level. The compensating behavior of task managers (partial maximization in the small) is not able to offset the fact that country managers fail to maximize at the global level.

There is good news and bad news in these results. The good news is that our analysis points to areas of improvement that could yield substantial benefits for the development impact of World Bank lending and the total resource transfer that the World Bank provides to developing countries. The bad news is that our



analysis of behavior suggests that in the past managers have had a strong interest in the level of commitments. This suggests a need for action. The World Bank should continue to strengthen the reward system such that managers pay even more attention to development impact. And it should educate managers about the importance of ESW in improving project quality.

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